

# Sriman Pankaj Boindala, PhD

Alexander von Humboldt Research Fellow — Expert in Water Systems Optimization & Resilience  
Chair of Digital Water Systems, Technical University of Berlin (TU Berlin), Germany  
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## SUMMARY

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Alexander von Humboldt Research Fellow specializing in the design, algorithmic optimization, and resilient management of urban water distribution systems under uncertainty. Proven expertise in integrating algorithmic fairness, socio-economic equity, and contaminant transport kinetics into water infrastructure modeling frameworks. Serves as an Associate Editor for ASCE's *Journal of Water Resources Planning and Management* and Chair of the Task Committee on GenAI in Publishing. Expert in deploying mathematical programming and optimization tools to ensure equitable resource allocation and infrastructure resilience.

**Research Interests:** Optimization and control of water distribution systems, water quality and engineering resilience modeling under uncertainty, equity/ fairness in Human-Water Infrastructure Systems, contaminant transport kinetics, and data-driven infrastructure management.

## EDUCATION

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- **Ph.D. Civil and Environmental Engineering**, Technion-Israel Institute of Technology, Haifa, Israel (2020–2023).  
*Thesis:* Uncertainty Inclusion in Design-Optimization of Multi-Quality Water Distribution Systems
- **M.Eng. Structural Engineering**, Birla Institute of Technology and Science, Pilani, India (2017–2019).  
*Thesis:* Investigation of Crushed Concrete Aggregates as a Replacement for Natural Aggregates in Alkali Activated Binder Concrete
- **B.Eng. Civil Engineering**, Birla Institute of Technology and Science, Pilani, India (2013–2017).  
*Thesis:* Understanding and Diagnosing Differences in Hydrological Model Structures

## EMPLOYMENT HISTORY

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**2026–Present: Alexander von Humboldt Research Fellow**, TU Berlin, Germany

- Project: *Adaptive Resilience of Water Distribution Systems - ARWADS*
- Researching water distribution networks, data science frameworks, and structural resilience under uncertainty at the Chair of Digital Water Systems.

**2025 Postdoctoral Researcher**, Technion-Israel Institute of Technology

- Project: *Israel-Germany Joint Research Grant - "StaySafe: Intermittent Agricultural Irrigation"*
- Collaborating with faculty from TU-Munich and Technion to develop numerical models for antimicrobial resistance gene (ARG) transport kinetics and bacterial regrowth in reclaimed wastewater irrigation systems.

**2024 Postdoctoral Researcher**, KIOS Research and Innovation Center of Excellence

- Project: *ERC Synergy Grant Water-Futures*
- Collaborated with KIOS, KWR Water Research Institute, Athens University, and Bielefeld University to develop robust optimization schedules for fair, equity-driven water quality and disinfection distribution in large-scale WDS.

## AWARDS

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- **Alexander Von Humboldt Research Fellowship (2026–2028)** - Alexander von Humboldt Foundation, Germany
- **Best Reviewer Award (2024)** - *Journal of Water Resources Planning and Management*, ASCE (\*Awarded at ASCE-EWRI Conference 2025)

- **Marie Skłodowska-Curie Actions – Seal of Excellence**, European Commission (2024 & 2025)  
Awarded to top-ranked proposals in the MSCA Individual Postdoctoral Fellowship program
- **Best Paper Award**, 2021 First International Conference on Recent Advancements in Civil Engineering (ICRACE 2021), NIT Silchar
- **Full Tuition Fee Waiver and Ph.D. Scholarship**, Technion-Israel Institute of Technology (2020–2023)
- **Best Oral Presentation Award**, 2018 National Symposium on Emerging Environmental Challenges (EEC 2018), CSIR-NEERI

## RESEARCH EXPERIENCE

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### Current Research Activities

#### *Israel-Germany Joint Research Grant - "Stay Safe for Intermittent Agricultural Irrigation"*

*Full-time Researcher*

**Focus:** Water Quality Modeling - Bacterial Growth, Transport, and Microbial Kinetics

- Studying bacterial regrowth and antimicrobial resistance gene transport within municipal and agricultural distribution networks.
- Developing numerical approximations to evaluate the fate of biological contaminants in recycled wastewater systems.

#### *ERC-Synergy Grant Water Futures Project – KIOS Research Team*

**Focus:** Algorithmic Fairness and Equity in Municipal Infrastructure Systems

- Collaborating with interdisciplinary partners to integrate social equity and distributive justice parameters into technical water networks.
- Developing advanced mathematical programming formulations to ensure fair and balanced disinfection levels across diverse municipal demographic sectors.

#### *Collaboration with INRS Canada and Technion*

*Self-initiated Collaboration*

**Focus:** Optimal Water Quality Management Considering Incomplete Mixing at Cross-Junctions

- Enhancing EPANET-IMX models to capture physical incomplete mixing uncertainties at cross-junctions, directly improving disinfectant concentration predictability.
- Designing robust optimization schedules for booster disinfection placement under structural and spatial uncertainty.

### Ph.D. Research - Technion-Israel Institute of Technology

#### *Uncertainty Inclusion in Design-Optimization of Multi-Quality Water Distribution Systems (WDS)*

- Developed advanced computational techniques to optimize water distribution networks under load, demand, and transport parameter uncertainties.
- Applied Evolutionary Algorithms, Robust Optimization, and Info-Gap Decision Theory in MATLAB/Python to optimize network pressure dynamics, maximizing structural reliability under severe uncertainty.
- Investigated solute mixing, disinfectant decay kinetics, and hydraulic transient variables to reinforce water infrastructure resilience.

## TEACHING & MENTORSHIP EXPERIENCE

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### Graduate Research Mentorship

- **Technical University of Berlin & Technion (2023–Present):** Formally and informally co-guided multiple PhD and Master's student researchers on mathematical modeling of contaminant kinetics, optimization routines, and hydraulic network analytics.

## Pedagogical Instruction & Course Support

- **Structural Optimization (CE G514)** – Teaching Assistant (Instructor: Prof. Vasan Arunachalam)  
Supported first-year Master’s students with numerical optimization assignments and MATLAB simulation projects.
- **Fluid Mechanics (CE F212)** – Teaching Assistant (Instructor: Prof. K. Srinivasa Raju)  
Authored a specialized engineering reference manual embedding MATLAB solutions for fluid mechanics and hydraulic network dynamics problems.
- **Introduction to Environmental Studies (BITS F224)** – Teaching Assistant  
Guided undergraduate capstone projects, structured evaluation matrices, and mentored student groups on sustainability concepts.
- **Hydrology (CE F321)** – Teaching Assistant (Instructor: Dr. Murari R. R. Varma)  
Formulated instructional coursework materials, project guidelines, and rubrics for surface water hydrology concepts.
- **Structural Dynamics (CE G551)** – Teaching Assistant  
Instructed graduate lab sessions focused on modeling computational structural response using code implementations.

## TECHNICAL & COMPUTATIONAL CAPABILITIES

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- **Water Systems Simulation:** EPANET, EPANET-IMX (Incomplete Mixing Model), WNTR, EPyT.
- **Optimization & Solvers:** Robust Optimization, Info-Gap Decision Theory, GUROBI, CVXPY, LINGO, Evolutionary Programming.
- **Programming & Spatial Data:** Python, MATLAB, R, ArcGIS, AutoCAD.

## INTERNATIONAL LEADERSHIP & PROFESSIONAL SERVICE

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### Editorial Roles

- **Associate Editor**, *Journal of Water Resources Planning and Management*, ASCE
- **Lead Guest Editor**, Special Collection on *Machine Learning and Artificial Intelligence in Water Distribution Systems*, *Journal of Water Resources Planning and Management*, ASCE
- **Co-Guest Editor**, Special Issue on *Optimisation of Water Distribution System Design and Operation Under Uncertainty*, *Water*, MDPI

### Task Committee Leadership

- **Chair**, ASCE-EWRI Task Committee on Generative AI in Publishing  
Leading international experts in formulating ethical guidelines and implementation policies for advanced generative models in scientific literature.
- **Member**, ASCE-EWRI Task Committee on ML/AI Applications in Water Engineering.

### Peer Review Activities

- **Journals:** *Journal of Water Resources Planning and Management* (ASCE), *Water Resources Management* (Springer), *Structural and Multidisciplinary Optimization* (Springer), *Urban Water Journal* (Taylor & Francis), *Hydroinformatics* (IWA), *Water* (MDPI).
- **Conferences:** ASCE EWRI World Environmental and Water Resources Congress (**2022, 2023, 2026**).

## PUBLICATIONS

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Google Scholar Citations: 202

h-index: 10

i10-index: 10

(As of March 2026)

## Journal Publications

1. **Boindala, S. P.**, Vrachimis, S. G., Eliades, D. G., & Polycarpou, M. (2025). Fairness in the distribution of water quality in drinking water supply systems. *Water Resources Management*. (\*\*Submitted). **Q1, SCIE**. (Focus: Socio-Hydrology & Equity).
2. Salfety, O., Sarne, O., **Boindala, S. P.**, Abhijith, G. R., & Ostfeld, A. (2025). A Cell Model for Pollutant Transport Quantification in Rainfall–Runoff Watershed Events. *Water*, 17(11), 1693. **Q1, SCIE**.
3. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2025). Booster Disinfection Scheduling under Uncertainty in Water Distribution Systems: Approximate Robust Reformulation Approach. *Journal of Water Resources Planning and Management*. **Q1, SCIE**.
4. **Sriman Pankaj, B.**, Vasana, A., & Jabez Christopher, J. (2024). Bio-inspired mix design optimization of self-compacting concrete using machine learning algorithms. *IJMMNO*. **Q4, Scopus**.
5. Abhijith, G. R., Naidu, M. N., **Boindala, S. P.**, Vasana, A., & Ostfeld, A. (2023). Analyzing the role of consumer behavior in coping with intermittent supply in water distribution systems. *Journal of Hydroinformatics*, 25(5), 1766–1787. **Q2, SCIE**.
6. Chundi, V., Raju, S., **Boindala, S. P.**, & Swain, S. S. (2023). A Non-Destructive Pavement Evaluation for the Development of a Multi Distress Regression Model Integrated with the BACKCSA Model. *IJPCRT*, 16(4), 873–887. **Q3**.
7. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2023). Robust Optimal Booster Disinfectant Injection in Water Systems under Uncertainty. *Water*, 15(9), 1777. **Q1, SCIE**.
8. Vogeti, R. K., **Boindala, S. P.**, Kumar, D. N., & Raju, K. S. (2022). Streamflow forecasting in a climate change perspective using E-FUSE. *Journal of Water and Climate Change*, 13(11), 3934–3950. **Q2, SCIE**.
9. **Pankaj, B. S.**, Jaykrishnan, G., & Ostfeld, A. (2022). Optimizing water quality treatment levels for water distribution systems under mixing uncertainty at junctions. *Journal of Water Resources Planning and Management*, 148(5), 04022013. **Q1, SCIE**.
10. **Boindala, S. P.**, & Ostfeld, A. (2022). Robust Multi-Objective Design Optimization of Water Distribution System under Uncertainty. *Water*, 14(14), 2199. **Q1, SCIE**.
11. Vasana, A., Srinivasa Raju, K., & **Sriman Pankaj, B.** (2022). Fuzzy optimization-based water distribution network design using self-adaptive cuckoo search algorithm. *Water Supply*, 22(3), 3178–3194. **Q2, SCIE**.
12. **Pankaj, B. S.**, Naidu, M. N., Vasana, A., & Varma, M. R. R. (2020). Self-adaptive cuckoo search algorithm for optimal design of water distribution systems. *Water Resources Management*, 34(10), 3129–3146. **Q1, SCIE**.
13. Raychoudhury, T., **Boindala, S. P.**, & Kalidindi, S. (2017). Performance evaluation of metal impregnated activated carbon composite for removal of fluoride. *Water Science and Technology: Water Supply*, 17(5), 1377–1385. **Q2, SCIE**.
14. **Pankaj, B. S.**, & Anmala, J. (2019). Investigation of latest techniques in carbon sequestration with emphasis on geological sequestration. *MOJ Eco Environ Sci*, 4(1), 7–12.

## Selected Refereed Conference Proceedings (Water Systems & Optimization)

1. **Boindala, S. P.**, Vrachimis, S. G., Eliades, D. G., & Polycarpou, M. (2025). Optimal Chlorine Dosage Scheduling in Drinking Water Distribution Systems Considering Fairness Objectives. *World Environmental and Water Resources Congress 2025*, 987–1000.
2. **Boindala, S. P.**, Perelman, G., & Ostfeld, A. (2025). Robust Sample Average Approximation for Optimal Chlorine Disinfection under Multiple Uncertainties. *World Environmental and Water Resources Congress 2025*, 1001–1009.
3. **Boindala, S. P.**, Yousefian, R., Duchesne, S., & Ostfeld, A. (2024). Robust Booster Disinfection Scheduling Using Incomplete Mixing Water Quality Model (EPANET-IMX). *World Environmental and Water Resources Congress 2024*, 1367–1379.

4. Vizanko, B., Shmaya, T., **Boindala, S. P.**, Ostfeld, A., & Berglund, E. (2024). Operating Water Distribution Systems for Equitable Access to Clean Water. *World Environmental and Water Resources Congress 2024*, 1229–1235.
5. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2023). Optimal Booster Chlorination Scheduling in WDS under Uncertainty: A Robust Counterpart Approach. *World Environmental and Water Resources Congress 2023*, 952–962.
6. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2022). Robust Multi-Objective Optimization of Water Distribution Systems. *World Environmental and Water Resources Congress 2022*, 1066–1075.
7. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2022). Source Treatment Level Optimization in Water Distribution Networks Considering Mixing Uncertainty. *World Environmental and Water Resources Congress 2022*, 1085–1095.
8. **Boindala, S. P.**, Abhijith, G. R., Ihjas, K., & Ostfeld, A. (2024). Towards Optimal Scheduling of Intermittent Water Supply Systems Incorporating Consumer Behavior. *Engineering Proceedings*, 69(1), 168.
9. **Boindala, S. P.**, & Ostfeld, A. (2024). Multiobjective optimization of water distribution system under uncertainty using robust optimization. *Proc. 2nd Joint CCWI/WDSA Conference*, Editorial Universitat Politècnica de València.
10. **Boindala, S. P.**, Jaykrishnan, G., & Ostfeld, A. (2024). Optimization of reservoir treatment levels considering uncertainty in mixing at cross junctions using Info-gap decision theory. *Proc. 2nd Joint CCWI/WDSA Conference*, Editorial Universitat Politècnica de València.
11. **Boindala, S. P.**, Abhijith, G. R., Aumeier, B. M., Steger, A., Salomons, E., Elkayam, R., Drewes, J. E., & Ostfeld, A. (2025). Challenges in Mathematical Modelling of Anti-Microbial Resistance Fate in Water Systems. *Proceedings of 21st International CCWI Conference, University of Sheffield*.
12. **Boindala, S. P.**, Abhijith, G. R., Ihjas, K., & Ostfeld, A. (2025). Equitable Water Access in Systems with Intermittent Supply. *Proceedings of 21st International CCWI Conference, University of Sheffield*.
13. **Boindala, S. P.**, Perelman, G., & Ostfeld, A. (2025). Optimal Booster Disinfection Scheduling Considering Demands and Disinfection Reaction Rate Uncertainty. *Proceedings of 21st International CCWI Conference, University of Sheffield*.
14. Wéber, R., Sándor, L., Abhijith, G. R., **Boindala, S. P.**, & Ostfeld, A. (2025). CFD Simulations of Incomplete Mixing at Pipe Junctions: Junction Angle Influence and Predictive Modelling. *Proceedings of the AMR in Water Systems Workshop, University of Sheffield*.
15. **Boindala, S. P.**, Yousefian, R., Robles, J., Duchesne, S., & Ostfeld, A. (2025). Optimizing Booster Chlorination in Water Distribution Networks Under Incomplete Mixing Conditions. *Proceedings of 21st International CCWI Conference, University of Sheffield*.
16. **Boindala, S. P.**, Arunachalam, V. (2019). Optimal Irrigation Planning and Operation of Reservoir using SACS. *3rd World Irrigation Forum (WIF3)*, 1–7.
17. Naidu, M. N., **Boindala, S. P.**, Vasan, A., & Varma, M. R. R. (2020). Optimization of water distribution networks using cuckoo search algorithm. *AEOTIT 2018*, 67–74.

#### Other Refereed Conference Proceedings

1. **Boindala, S. P.**, Ramagiri, K. K., & Kar, A. (2022). Investigation of Crushed Concrete Aggregates as a Replacement for Natural Aggregates in AAB Concrete. *ICRACE 2021*, 55–63.
2. Kar, A., Ramagiri, K. K., **Boindala, S. P.**, Ray, I., Halabe, U. B., & Unnikrishnan, A. (2022). Investigations on Chemical, Mechanical, and Long-Term Characteristics of Alkali-Activated Concrete. *Advances in Sustainable Materials and Resilient Infrastructure*, 133–144.
3. Ramagiri, K. K., **Boindala, S. P.**, Zaid, M., & Kar, A. (2021). Random forest-based algorithms for prediction of compressive strength of ambient-cured AAB concrete. *International Conference on Structural Engineering and Construction Management*, 717–725.

4. **Boindala, S. P.**, Arunachalam, V. (2020). Concrete mix design optimization using a multi-objective cuckoo search algorithm. *SoCTA 2018: Soft Computing: Theories and Applications*, 119–126.
5. **Boindala, S. P.**, Ramagiri, K. K., Alex, A., & Kar, A. (2020). Step-Wise Multiple Linear Regression Model Development for Shrinkage Strain Prediction of Alkali Activated Binder Concrete. *SECON'19*, 91–98.

## INVITED TALKS

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- **Boindala, S. P.** (November 2025). “Steady-State and Extended Period Simulations (EPS): Understanding Pressures, Flows, and Demands”, *Short-Term Course on Source to Tap: Design and Analysis of Water Supply Systems*, Indian Institute of Technology (IIT) Kanpur.

## REFERENCES

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